MODULE 6 · OPERATIONS AND HUMAN FACTORS

Lesson 27: CRM and Threat and Error Management in Winter Operations

Lesson 27 of 30 · about 8 min

LEARNING OBJECTIVES

- Identify typical winter threats and errors.
- Apply TEM countermeasures, including briefings and pre-agreed decision points.
- Recognise and resist plan continuation bias and commercial pressure.



Shared monitoring: Both pilots actively cross-check aircraft configuration, de-icing status, holdover time, and runway/contamination reports.
Clear roles: Assign who calls HOT start/expiry, who performs the final contamination check, and who communicates with ground services.
Unambiguous communication: Use concise, standardized callouts with ATC, dispatch, ramp crews, and de-icing teams to avoid fluid or timing errors.
Confirm and agree: During pre-takeoff checks both pilots must verbalize findings, confirm checklist completion, and jointly declare the aircraft "clean and safe" before departure.
Conservative decision making: If uncertainty exists, re-inspect, re-treat, or delay departure—never accept ambiguity.

Shared monitoring and clear decision points protect the crew from time pressure.

Typical winter threats

- Changing precipitation type and intensity; darkness and low visibility;
- Time pressure: slot times, passenger connections, crew duty limits, de-icing queues;
- Complex communication between ATC, pad control, the de-icing crew and the cabin;
- Unfamiliar procedures at airports that rarely have winter weather;
- Fatigue after early starts and long delays;
- Equipment issues: unserviceable APU, limited de-icing equipment.

Typical errors

- Using the wrong HOT table, fluid, concentration or start time;
- Not updating the HOT when conditions change;
- Skipping or rushing the winter-specific checklist items: anti-ice, run-ups, flaps after slush taxi, flight-control check;
- Using reduced thrust or wrong performance data on a contaminated runway;

- Assuming the de-icing crew treated an area — or assuming the flight crew is ready.

Countermeasures

1. **Winter briefing.** Weather, runway conditions, de-icing plan, HOT strategy, performance, anti-ice, taxi route and hazards.
2. **Pre-agreed decision points.** “If HOT expires before take-off, we return.” “If RWYCC falls below 3, we wait or divert.” Decisions made calmly before the pressure arrives are better than decisions made under it.
3. **Monitoring and cross-checking.** Both pilots calculate the HOT independently; the pilot monitoring tracks time and conditions.
4. **Speaking up.** Any crew member who sees a threat must say so clearly. Use the company's assertiveness model — for example: get attention, state the concern, propose an action, get agreement.
5. **Workload management.** Do the de-icing and HOT tasks at a planned moment, not during the taxi or take-off briefings.

Plan continuation bias

The closer we get to completing a plan, the harder it is to change it — even when the information shows we should. In winter this appears as “just one more aircraft ahead of us, the HOT will be fine”, or “the runway was fine for the last aircraft”. Recognise it, name it, and go back to your pre-agreed decision points.

> **ORGANISATIONAL SUPPORT**
> Domestic Airlines supports any crew decision to delay, return for de-icing or divert for safety reasons. No commercial consideration justifies taking off with a contaminated aircraft.

KEY TAKEAWAYS

- Brief winter threats and agree decision points before the pressure starts.
- Both pilots calculate HOT; the pilot monitoring tracks time and conditions.
- Recognise plan continuation bias — go back to your decision points.

KNOWLEDGE CHECK

Which countermeasure best protects a crew from time pressure at the holding point?

A. Taxiing faster

B. A pre-agreed decision point, briefed before taxi, for when HOT expires

C. Skipping the HOT calculation to save time

D. Asking ATC for a shorter route

Answer: B. Decision points agreed calmly before taxi remove the need to decide under pressure and help resist plan continuation bias.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.